BLMRS: A Balanced Learning Method for Multimodal Remote Sensing Image Classification

Qian Liu, Feng Cao, Tianjun Wu, Deyu Li, Yuhua Qian, *Member, IEEE*, and Xia Zhang

***Abstract*—The multimodal remote sensing image classification provides essential support for fine-grained understanding of the Earth's surface by fusing the complementary information. Yet, the multimodal fusion often suffers from a modal imbalance problem, where applying a uniform optimization objective leads to the under-optimization of certain modalities, posing a bottleneck to realizing the full benefits of fusion. Existing balanced learning strategies attempt to solve this problem by emphasizing the training of encoders for weaker modalities. However, in remote sensing scenarios, they overlook the domain-specific reality that certain weaker modalities inherently lack discriminative information; for example, the LiDAR-derived digital surface model (DSM), which does not provide the necessary detail to effectively distinguish between land cover types. Indiscriminately emphasizing the learning of such modalities may introduce noise and impair the model's overall performance. To address this, this work proposes a balanced learning method for multimodal remote sensing image classification (BLMRS). The method takes unimodal training in a noncompetitive environment as a reference and uses its output as the guidance signals to dynamically adjust the optimization preference in multimodal learning. This allows adaptive optimization for weaker modalities, avoiding overemphasis when their discriminative information is limited. To further avoid conflicts among guidance signals from different unimodal sources, we measure and filter out components inconsistent with the weaker modality's guidance, ensuring the training focus remains on it. BLMRS is architecture-agnostic and can be seamlessly integrated into existing classification backbones. Extensive experiments on four public benchmarks and six backbone architectures demonstrate that our method consistently offers performance gains. It remains robust even when a modality's discriminative information is degraded.**

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***Index Terms*—Image classification, modal imbalance problem, multimodal fusion, remote sensing.**

I. Introduction

With the advancement of remote sensing sensor technologies and the diversification of data acquisition methods, remote sensing images exhibit increasingly pronounced multimodal characteristics [1], [2]. By integrating the multisource data, such as optical imagery, synthetic aperture radar (SAR), and digital surface model (DSM), we can overcome the physical limitations of single-type sensors, thereby significantly improving the accuracy and robustness of remote sensing image classification [3], [4], [5].

Due to differences in resolution, imaging mechanisms, and acquisition conditions among various remote sensing modalities [1], the amount of task-relevant information provided by each modality varies [6], [7]. This variation leads to the modal imbalance problem when jointly training multiple modalities [8], [9], where certain modalities learn significantly less than others [10], [11], [12], [13], [14], or all modalities fail to reach sufficient optimization within the multimodal model [15]. The problem restricts the potential of multimodal fusion, resulting in the suboptimal performance in practical applications [16]. In recent years, several efforts have been devoted to addressing this problem [8], [16], [17], [18], [19], [20], [21]. The predominant strategy among these efforts involves evaluating the learning quality of each modality based on its unimodal predictive accuracy within the multimodal model, then dynamically adjusting the fusion process by emphasizing the training of encoders for weaker performing modalities while suppressing dominant ones [16], [17], [18], [20], [21].

However, although this strategy shows promise in general multimodal learning tasks, it falters when applied to remote sensing. The reason is that it overlooks a domain-specific reality: the discriminative information carried by certain remote sensing modalities is inherently limited. A typical example is the DSM, which encodes the elevation information in a single channel and often lacks sufficient semantic cues for distinguishing land cover types. For instance, as shown in Fig. 1, classes like *soil* and *healthy grass* can exhibit completely overlapping elevation values, causing distinct classes to become indistinguishable from the DSM's perspective. For a modality that is poor in discriminative information, the learning quality it can support is inherently bounded, and its potential contribution to fusion is bounded as well. Therefore, even if such a modality performs weaker in multimodal

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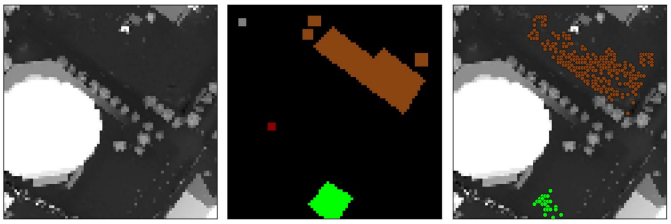


Fig. 1. Illustration of limitations of low-discriminative information modalities using the Houston 2013 dataset [22]. (a) DSM image, (b) ground-truth labels (brown: *soil* and green: *healthy grass*), and (c) samples with identical DSM elevation values but belonging to different classes, showing that these classes are difficult to distinguish from the DSM perspective due to limited discriminative power. See Part I of the supplementary material for more examples of conflicts.

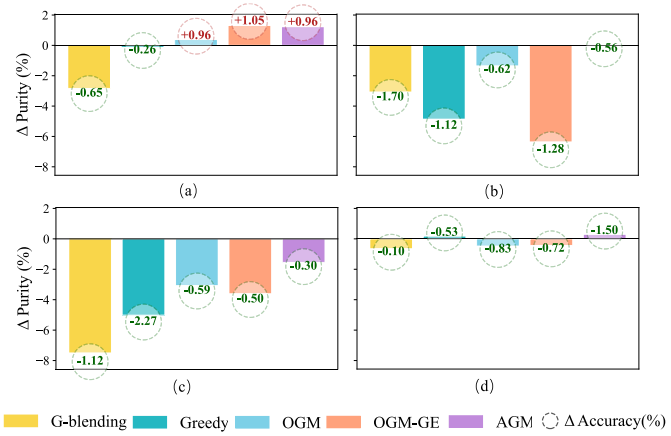


Fig. 2. Performance of balanced learning methods as the DSM discriminative information is progressively degraded on Houston 2013. The compared balanced learning methods (G-Blending [16], Greedy [20], OGM [17], OGM-GE [18], and AGM [21]) are described in detail in Section II-B. Under these low-discriminative conditions, the classification accuracy becomes unreliable, so we assess modality learning quality with a feature separability metric instead, where separability is measured in terms of cluster purity. Bar height indicates whether unimodal feature separability improves or degrades compared to the baseline. Numbers inside circles show changes in the multimodal classification accuracy: *red dashed circles* indicate improvement and *green circles* indicate decline. See Section IV-H for details on discriminative information reduction. (a) Raw elevation resolution. (b) Moderately degraded elevation. (c) Severely degraded elevation. (d) Extremely degraded elevation.

learning, this does not always mean that it is insufficiently optimized during joint training. Indiscriminately emphasizing their learning does not bring additional benefits; instead, it may introduce noise that degrades the overall performance of the multimodal model. To empirically validate this principle, we simulate the degradation of discriminative information by reducing the elevation resolution of the DSM. The results are presented in Fig. 2. Although some methods demonstrate effectiveness at the raw resolution, their performance drops markedly when the discriminative information of the DSM is further reduced, ultimately failing to improve the learning quality of this modality and the overall model performance. This phenomenon underscores the unsuitability of this strategy for a remote sensing scene.

Beyond the limitations of low-discriminative modalities, the architectural diversity of multimodal remote sensing networks also poses challenges for the adaptability of balancing strategies. This diversity stems from the heterogeneous and multiscale characteristics of remote sensing data, which have driven the development of specialized network architectures, including multistage convolutional fusion, attention-based models, and hierarchical designs [5], [23], [24]. However, many existing balanced learning strategies are tightly coupled with specific architectural choices, such as classification heads with separable weights or explicit distinction between the parameters of the encoder for different modalities, to facilite estimate per-modality learning quality or to balance optimization [11], [16], [17], [18], [19], [20], [21], which limits their portability across different fusion models.

To address this, we propose balanced learning method for multimodal remote sensing image classification (BLMRS), which treats the unimodal performance measured under a noncompetitive environment, where each modality learns independently and is without interference by cross-modal competition, as a reference of that modality's information capacity, and leverages this reference to balance the multimodal learning dynamically in an architecture-agnostic manner. Specifically, we proposed a modality state-aware (MSA) strategy that guides the weight allocation by defining each modality's learning state from the perspective of feature separability in the feature space. This state is determined by two factors: modality dependency and its optimization potential. The former is quantified by the performance shift when the modality is removed or added, reflecting this modality's relative dominance or weakness. The latter is measured by the gap between the modality's performance within the multimodal model and its unimodal reference, indicating whether there is room for further improvement. Based on this weight, BLMRS adopts a teacher–student-like unimodal guidance mechanism, where the outputs of unimodal networks training without cross-modal interference are used to adjust the multimodal model's learning preferences, thus alleviating the modal imbalance problem. However, when multiple unimodal networks are involved simultaneously, inconsistent guidance signals may emerge, leading to conflicts, we design a Kendall rank correlation (KRC)-based [25] consistency metric to filter out contradictory guidance, ensuring stable and balanced learning.

Our main contributions are fourfold.

1) Introduce a unimodal guidance mechanism leveraging unimodal networks training in a noncompetitive setting as references, enabling the learning process to focus on the weaker modality while avoiding overemphasis.
2) Propose an MSA strategy that estimates the learning state of each modality based on the feature separability, supporting balanced learning processing to adjust multimodal model learning preference dynamically.
3) Design a KRC-based consistency metric to filter conflicts among multiple unimodal network guidance signals, ensuring the continuous attention to weaker modalities during the balanced learning process.
4) Our method is architecture-agnostic, which relies solely on feature representations and model outputs for adjusting balanced learning, making it a plug-and-play module for multimodal remote sensing classification models.

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II. Related Works

This section first reviews studies on the multimodal remote sensing image classification and then reviews the main methods for addressing the modal imbalance problem, and further analyzes its causes.

A. Multimodal Remote Sensing Image Classification

To improve the classification performance of multimodal remote sensing data, researchers have explored diverse deep-learning architectures, including CNN-, Transformer-, and Mamba-based models. Initially, early work mainly adopted CNN structures, such as two-branch or coupled CNNs, to extract and fuse features from hyperspectral (HS) imagery and DSM data [3], [26]. With the rise of Transformer architectures, their strong ability to model long-range dependencies motivated their adoption in multimodal remote sensing classification tasks [5], [23], [27], [28]. However, despite their effectiveness in capturing global contextual information, exploiting fine-grained local spatial cues remains challenging [29]. To alleviate this limitation, several variants refine Transformer blocks to enhance local interactions, exemplified by SpectralFormer [30], while ExViT further strengthens multimodal fusion through parallel branches and cross-modal attention mechanisms [24]. Most recently, the Mamba architecture [31] has been tailored to HS data [33], [34], and has further been introduced into multimodal remote sensing models, where its efficient long-range sequence modeling helps to improve the overall classification performance [32].

Nevertheless, even with improved intermodal interaction, differences in modality information content continue to hinder effective fusion [35], [36], keeping imbalanced learning an open challenge [8].

B. Balanced Learning Methods

In recent years, the imbalanced learning problem has attracted increasing attention, leading to the development of various solutions. Existing methods primarily focus on the gradient modulation, including employing gradient-mixing techniques to balance the learning rates of different modalities during optimization [17], [18], [37], and adopting real-time gradient modulation mechanisms to dynamically adjust the contribution ratio of each modality for more balanced training [21]. Some studies also highlight that optimization effectiveness depends not only on gradient magnitude but also on its direction. Based on this, strategies such as analyzing gradient conflicts, enforcing alignment between multimodal and unimodal gradients, and enhancing their magnitudes have been proposed to mitigate the modal imbalance [38]. However, such approaches typically rely on explicit encoder control or separable fusion weights, which limit their general applicability [11], [21]. Beyond gradient-level approaches, how to adjust training strategies to achieve balanced multimodal learning has also attracted the attention of researchers recently. For example, modality curriculum learning reduces performance gaps by training modalities with large disparities in a staged manner [12]; alternating optimization strategies train different modality encoders in turn [39], achieving notable results but further increasing the dependency on explicit encoder designs.

In the field of remote sensing, investigations into the modal imbalance problem are still very limited. A few task-specific attempts exist—for instance, a study leveraging LiDAR–aerial imagery complementarity with attention and network optimization for land cover segmentation [9], and another employing consistency constraints and dual optimization schemes for building extraction [8]. However, systematic research on the modal imbalance problem in multimodal remote sensing image classification remains scarce, and no mature and effective solutions have been proposed.

C. Analysis

We analyzed the modal imbalance problem and showed that learning conflicts and competition arise between different modalities during the multimodal fusion process. This leads to inconsistent gradient updates, preventing some modalities from being fully learning. To facilitate the analysis, we adopt a simplified linear fusion model.

Given a training sample $(x_i^{(m1)}, x_i^{(m2)}, \mathbf{y}_i)$, where $x_i^{(m1)}$ and $x_i^{(m2)}$ are inputs from two modalities and $\mathbf{y}_i \in \{0,1\}^K$ is the one-hot ground-truth label, where K is the number of ground-truth classes, and the multimodal predicted probability vector is formulated as follows:

$$\mathbf{p}_i^{(f)} = \sigma\left(\left[e^{(m1)}\left(x_i^{(m1)}\right), e^{(m2)}\left(x_i^{(m2)}\right)\right]W + b\right) \tag{1}$$

where $e^{(m1)}(\cdot)$ and $e^{(m2)}(\cdot)$ denote the encoders, $W$and b are the classifier parameters, and σ is the softmax function. The objective is to minimize the cross-entropy loss

$$\mathcal{L}^{(f)} = -\frac{1}{N}\sum_{i=1}^{N}\mathbf{y}_i^{\top}\log\mathbf{p}_i^{(f)} \tag{2}$$

where $\mathcal{L}^{(f)}$ denotes the multimodal loss and N is the number of training set. During multimodal training, parameters $\theta^{(m1)}$ of modality $m1$ are updated via gradient descent

$$\theta^{(m1)} \leftarrow \theta^{(m1)} - \eta\frac{\partial\mathcal{L}^{(f)}}{\partial\theta^{(m1)}} \tag{3}$$

where η is the learning rate, with the gradient

$$\frac{\partial\mathcal{L}^{(f)}}{\partial\theta^{(m1)}} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{\partial e^{(m1)}\left(x_i^{(m1)}\right)}{\partial\theta^{(m1)}}\right)^{\top} W^{(m1)^T}\left(\mathbf{p}_i^{(f)} - \mathbf{y}_i\right) \tag{4}$$

where N is the number of samples, $W^{(m1)}$ refers to the part of W associated with modality $m1$ [17], [18], and the detailed derivation is given in Part II of the supplementary material.

In the unimodal case (only $m1$), the predicted probability vector become: $\mathbf{p}_i^{(m1)} = \sigma(e^{(m1)}(x_i^{(m1)})W^{(m1)} + b)$, and the gradient is

$$\frac{\partial\mathcal{L}^{(m1)}}{\partial\theta^{(m1)}} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{\partial e^{(m1)}\left(x_i^{(m1)}\right)}{\partial\theta^{(m1)}}\right)^{\top} W^{(m1)^T}\left(\mathbf{p}_i^{(m1)} - \mathbf{y}_i\right) \tag{5}$$

where $\mathcal{L}^{(m1)}$ represents the cross-entropy loss in unimodal setting.

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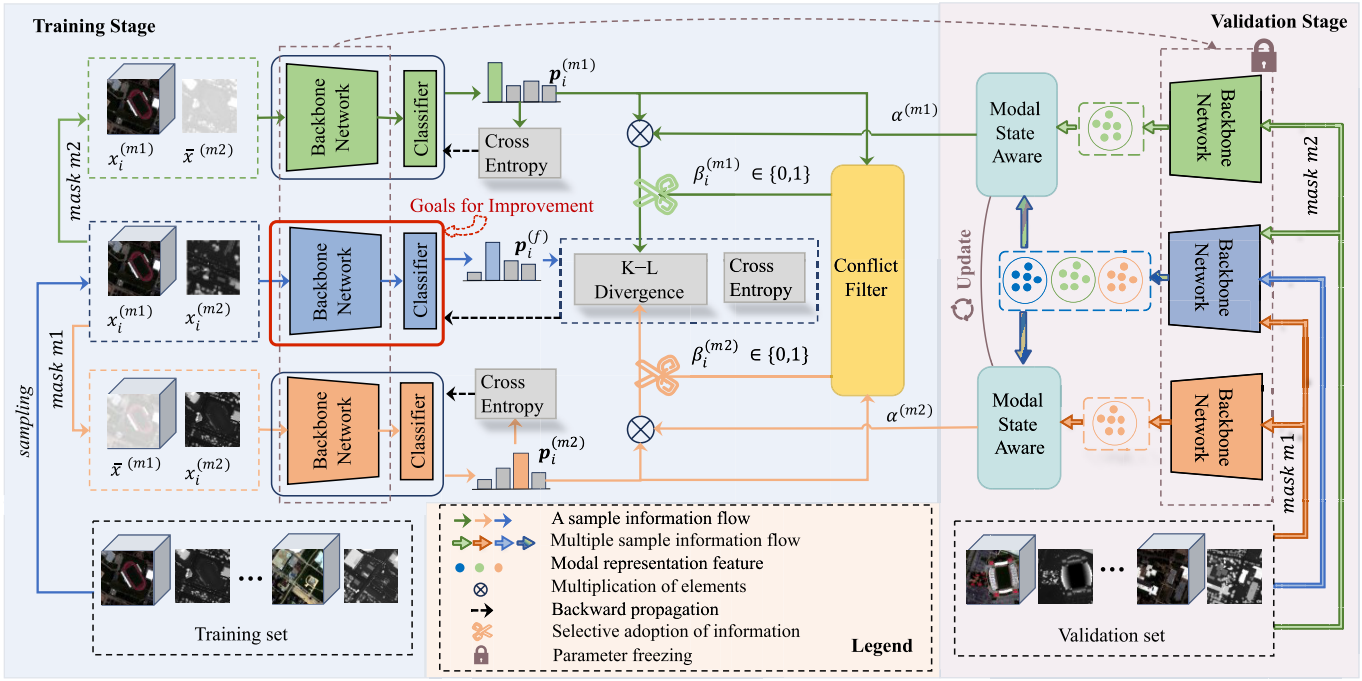


Fig. 3. Illustration of the BLMRS method. The models within the black boxes represent unimodal networks, while those within the red boxes denote the multimodal model, which is the target of improvement. During training, unimodal inputs are constructed by applying specific modality masks to the unimodal data. The multimodal model is optimized not only by using classification loss but also by using the KL divergence to correct its learning. The conflict filter selects reliable guidance signals based on the consistency between the outputs of the unimodal networks. In the validation stage, all validation samples are processed collectively to compute and update the MSA values, which guide subsequent training.

By comparing (4) and (5), it can be observed that although both multimodal and unimodal models are optimized toward the same ground-truth label $\mathbf{y}_i$, the probability vector from the fused model $\mathbf{p}_i^{(f)}$ often deviates from its unimodal counterpart $\mathbf{p}_i^{(m1)}$ [10]. Since the gradient for the encoder of modality $m1$ is determined by $\mathbf{p}_i^{(f)} - \mathbf{y}_i$ in multimodal learning, such deviation causes a misalignment with the modality-specific signal $\mathbf{p}_i^{(m1)} - \mathbf{y}_i$, leading to conflicting update. The larger this discrepancy, the more difficult it becomes for the modality to follow its optimal learning trajectory, thereby hindering feature learning and degrading the quality of the fused representation. This also indicates that, within a multimodal learning framework, all modalities are subject to varying degrees of interference. In cases of severe imbalance, the dominant modality largely determines $\mathbf{p}_i^{(f)}$, suppressing the effective learning of others [18]. Thus, we think that the key to solving this problem is how to reduce information bias in the learning process, thereby promoting balanced learning across all modalities.

III. Method

In this section, we propose the BLMRS method, which overall framework is shown in Fig. 3. It follows a sequential pipeline: it first estimates modality learning states, then adjusts multimodal learning accordingly, and finally filters conflicting guidance signals to ensure the stability and balance. This pipeline is realized through three key components: 1) an MSA strategy; 2) a unimodal network guidance mechanism; and 3) a KRC-based consistency metric. Sections III-A–III-C detail the design and implementation of each component.

A. Modality State Aware

This section proposes the MSA strategy, as illustrated in Fig. 4. It uses feature separability to estimate the learning state of each modality and provide guidance weights for the subsequent balancing process. Taking the modality m as an example, we evaluate each modality learning state from two perspectives.

1) *Degree of Modality Dependency:* The performance change resulting from adding or removing modality m, while keeping others fixed, reflects its relative informational contribution. A larger change indicates that modality m provides more information and can be regarded as a dominant modality. Conversely, a smaller change suggests that modality m plays a weaker role and is more likely to experience insufficient optimization during training.
2) Whether the modality retains further optimization potential. We assess this by comparing the performance of a modality in multimodal joint training with that in a unimodal setting without cross-modality competition. If the performance of a modality in the multimodal setting is already close to its unimodal optimal level, it indicates that already sufficiently emphasized during training or its information capacity is inherently limited, and further emphasis would yield minimal gains. Conversely, if there remains a significant gap, it suggests underoptimization, necessitating further refinement.

In order to quantify the above two metrics more reliably and to avoid the possible misleading nature of the accuracy when a modality carries low-discriminative information. We adopt

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feature separability, measured by purity, as the basis of evaluation [40]. Features with higher separability generally correspond to better learning quality and stronger discriminative capacity. Purity provides a direct and computationally simple way to quantify separability. Given a sample $\mathbf{x}_i$, we extract its feature representation from the backbone network: $f(\mathbf{x}_i)$, and apply k-means clustering over the feature set $\{f(\mathbf{x}_i)\}_{i=1}^{N_{val}}$, N_{val} is number of sample in MSA strategy, where the number of clusters is set to K. This produces a clustering result $O = \{O_1, \ldots, O_K\}$, where each O_k is a cluster containing a subset of samples. We compare these clusters with the ground-truth label partitions $G = \{G_1, \ldots, G_K\}$, where each G_j is the set of samples belonging to class j and compute the clustering purity as follows:

$$\tau = \frac{1}{N_{val}} \sum_{O_k \in O} \max_{G_j \in G} |O_k \cap G_j| \tag{6}$$

where $|O_k \cap G_j|$ denotes the number of samples in cluster O_k that belong to ground-truth class j, N_{val} represents the number of samples we want to cluster. A higher τ indicates better feature separability. We define four types of feature purity for subsequent computations.

1) $\tau^{f(mm)}$: Feature purity using all modalities as input.
2) $\tau^{f(-m)}$: Feature purity with modality m masked.
3) $\tau^{f(m)}$: Feature purity with only modality m active (others masked by mean features; see Section III-B).
4) $\tau^{(m)}$: Feature purity from the unimodal network trained solely on modality m.

Inspired by the concept of isolating unimodal responses, see [15], [41], [42], we define the dependency of modality m as follows:

$$c^{(m)} = \left\lfloor \tau^{f(mm)} - \tau^{f(-m)} \right\rfloor_0 \tag{7}$$

where $\lfloor \cdot \rfloor_0$ denotes clipping the minimum value to 0. To quantify whether a modality retains potential for further optimization, we define the $d^{(m)}$ as a measurement

$$d^{(m)} = \left\lfloor \tau^{(m)} - \tau^{f(m)} \right\rfloor_0 . \tag{8}$$

Finally, the modality state weight $\alpha^{(m)}$ is defined by combining $c^{(m)}$ and $d^{(m)}$ as follows:

$$\alpha^{(m)} = exp\left(1 - \frac{c^{(m)}}{\sum_{i=1}^{M} c^{(i)} + \epsilon}\right) \cdot d^{(m)}, \quad \alpha^{(m)} \geq 0 \tag{9}$$

where exp denotes the exponential function, ϵ is a small constant added to avoid division by 0, and M is the total number of modalities (in this work, $M = 2$). A larger value of $\alpha^{(m)}$ indicates that greater focusing should be placed on learning for the corresponding modality. In practice, we calculated the metric on the validation set, so the sample number is denoted by N_{val}. This choice ensures that MSA reflects the model's generalization behavior and reduces computational overhead. The modality scores $\alpha^{(m)}$ are updated once per validation epoch. The detailed implementation is provided in Algorithm 1.

Algorithm 1 MSA

Input: Validation set $\{(x_i^{(m1)}, x_i^{(m2)}), y_i\}_{i=1}^{N_{val}}$;
Multimodal backbone $f(\cdot)$; Unimodal backbones $f^{(m1)}(\cdot)$ and $f^{(m2)}(\cdot)$
Output: Modality state weigh $\alpha^{(m1)}$ and $\alpha^{(m2)}$
Extract feature representations from $f(\cdot)$ with both modalities active, compute feature purity $\tau^{f(mm)}$ via (6)
for each modality $m \in \{m1, m2\}$ **do**
Mask modality m and extract features using $f(\cdot)$, compute $\tau^{f(-m)}$ via (6)
Activate only modality m and extract features using $f(\cdot)$, compute $\tau^{f(m)}$ via (6)
Extract features using unimodal backbone $f^{(m)}(\cdot)$, compute $\tau^{(m)}$ via (6)
Compute modality dependency score $c^{(m)}$ via (7)
Compute optimization potential $d^{(m)}$ via (8)
Compute final modality state indicator $\alpha^{(m)}$ via (9)
end for
Return $\alpha^{(m1)}$ and $\alpha^{(m2)}$

B. Unimodal Network Guidance Mechanism

This section describes how to construct unimodal networks and leverage them to adjust the multimodal model's optimization preference toward a specific modality, with the aim of mitigating the modal imbalance problem.

1) Construction and Training of Unimodal Networks: Unlike the idea in traditional teacher–student network mechanism designs [43], [44], which use a powerful model as the teacher for guidance. Our method aims to ensure the balanced multimodal learning and avoid overestimating the multimodal model's performance. Therefore, we treat the backbone $f(\cdot)$ as a black-box function and construct a unimodal network by copying the architecture of $f(\cdot)$ and initializing them with the same parameters. Each unimodal network is constructed by masking nontarget modalities in the input. Specifically, for the unimodal network of modality $m1$, we replace the input of the other modality $m2$ with its mean vector computed over the entire training dataset: $\bar{x}^{(m2)} = 1/N \sum_{i=1}^{N} x_i^{(m2)}$. This operation removes information specific to the other modality while preserving the overall input distribution [21]. The probability of the unimodal network for modality $m1$ are computed as follows:

$$\mathbf{p}_i^{(m1)} = \sigma\left(h^{(m1)}\left(f^{(m1)}\left(x_i^{(m1)}, \bar{x}^{(m2)}\right)\right)\right) \tag{10}$$

where $h^{(m1)}$ is the classification head and $f^{(m1)}(\cdot)$ is the backbone of unimodal network.

Similarly, the loss function used in the unimodal network is kept consistent with that in the multimodal model. The cross-entropy loss for modality $m1$ is defined as follows:

$$\mathcal{L}^{(m1)} = -\frac{1}{N} \sum_{i=1}^{N} \mathbf{y}_i^{\top} \log \mathbf{p}_i^{(m1)}. \tag{11}$$

This loss guides the unimodal network to make modality-specific predictions aligned with the labels.

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Algorithm 2 Unimodal Network Guidance Mechanism

Input: Training dataset $D = \{(x_i^{(m1)}, x_i^{(m2)}), y_i\}_{i=1}^N$ with two modalities; Multimodal backbone $f(\cdot)$ and classification head $h^{(f)}$

Output: Trained multimodal model $(f, h^{(f)})$

for each modality $m \in \{m1, m2\}$ **do**
  Initialize unimodal backbone $f^{(m)} \leftarrow f$
  Initialize unimodal classification head $h^{(m)} \leftarrow h^{(f)}$
end for
for each training mini-batch $B \subset D$ **do**
  Compute multimodal predicted probability vector via (12)
  Compute multimodal cross-entropy loss $\mathcal{L}^{(f)}$ via (2)
  for each modality $m \in \{m1, m2\}$ **do**
    Replace the other modality's input with its mean vector to create masked input
    Compute unimodal predicted probability vector $\mathbf{p}_i^{(m)}$ via (10)
    Compute unimodal cross-entropy loss $\mathcal{L}^{(m)}$ and update $(f^{(m)}, h^{(m)})$
    Compute guidance loss $\mathcal{L}_{\text{kd}}^{(m)}$ via (13)
  end for
  Update multimodal model $(f, h^{(f)})$ using total loss: $\mathcal{L} = \mathcal{L}^{(f)} + \mathcal{L}_{\text{kd}}^{(m1)} + \mathcal{L}_{\text{kd}}^{(m2)}$
end for
Return: Trained multimodal model $(f, h^{(f)})$

2) Guiding the Learning Process of the Multimodal Model: We employ Kullback–Leibler (KL) divergence to guide multimodal learning using unimodal outputs. As KL divergence is applied at the output level [43], [44], it directly regularizes the probability distributions of the multimodal model against those of unimodal networks, providing a simple and architecture-agnostic advantage. For clarity and to facilitate correspondence with unimodal outputs, the predicted probability of the multimodal model in (1) is rewritten as follows:

$$\mathbf{p}_i^{(f)} = \sigma\left(h^{(f)}\left(f\left(x_i^{(m1)}, x_i^{(m2)}\right)\right)\right) \tag{12}$$

where $h^{(f)}$ is the classification head of the multimodal model. The guidance objective function for modality $m1$ in the multimodal model using KL divergence is defined as follows:

$$\begin{aligned}\mathcal{L}_{\text{kd}}^{(m1)} &= \alpha^{(m1)} T^2 \sum_{i=1}^{N} \text{KL}\left(\mathbf{p}_i^{(f)} \,\|\, \mathbf{p}_i^{(m1)}\right) \\ &= \alpha^{(m1)} T^2 \sum_{i=1}^{N} \mathbf{p}_i^{(f)} \log\left(\frac{\mathbf{p}_i^{(f)}}{\mathbf{p}_i^{(m1)}}\right)\end{aligned} \tag{13}$$

where T is the temperature factor (set to 2 in this work, following common practice in KL-based methods), $\text{KL}(\cdot\|\cdot)$ denotes the KL divergence function, and $\alpha^{(m1)}$ is the modality weight derived from the MSA. Similarly, for modality $m2$, the same procedure is performed using its dedicated unimodal network to produce predicted probability vector $\mathbf{p}_i^{(m2)}$, and computing the corresponding objective function $\mathcal{L}_{\text{kd}}^{(m2)}$ in the same manner as (13). Finally, during training, the multimodal model is optimized by a joint classification loss (cross-entropy loss) and guidance loss. The full procedure is summarized in Algorithm 2.

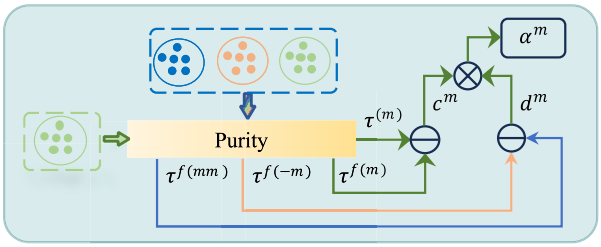


Fig. 4. Illustration of MSA strategy. Take the modality m as an example.

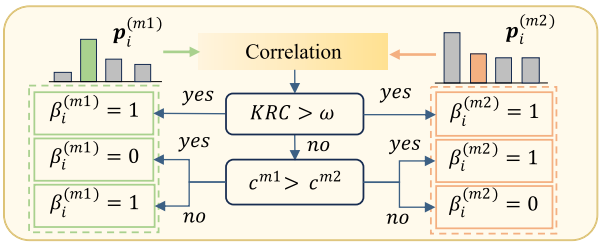


Fig. 5. Illustration of the process of using the KRC-based consistency metric for conflict filtering.

C. KRC-Based Conflict Filtering

Although the MSA dynamically guides the influence of unimodal networks, when substantial discrepancies exist between unimodal networks' predictions for the same input [45]. The resulting conflicting signals impose contradictory learning objectives on the multimodal model that hinder multimodal learning. To mitigate this issue, it is essential to assess the consistency of unimodal outputs before applying guidance. Therefore, we adopt the KRC to quantitatively evaluate the agreement between the predicted probability by different unimodal networks. This consistency measurement, as illustrated in Fig. 5, enables us to detect and address potential conflicts in guidance signals during training.

Let $p_{i,a}^{(m)}$ be the predicted probability of sample i belonging to class a under unimodal network m. Based on these probability distributions $\mathbf{p}_i^{(m1)}$ and $\mathbf{p}_i^{(m2)}$, the KRC coefficient is defined as follows:

$$\begin{aligned}\text{KRC}\left(\mathbf{p}_i^{(m1)}, \mathbf{p}_i^{(m2)}\right) = \frac{2}{K(K-1)} \sum_{a<b} \text{sign}\left(p_{i,a}^{(m1)} - p_{i,b}^{(m1)}\right) \\ \cdot \text{sign}\left(p_{i,a}^{(m2)} - p_{i,b}^{(m2)}\right)\end{aligned} \tag{14}$$

where $p_{i,a}^{(m1)}$ denotes the predicted probability of sample i belonging to class a under the unimodal network of modality $m1$, and other terms with the same form are defined analogously.

$\text{sign}(\cdot)$ is the sign function that returns $+1$ if the argument is positive, -1 if it is negative, and 0 if it is 0. The summation $\sum_{a<b}$ runs over all unordered pairs of distinct classes (a, b). The multimodal model uses both the KRC coefficient and the dependency score c^m to determine which unimodal network's guidance signals to adopt. Specifically, for samples where KRC < 0, we discard the guidance from the unimodal network corresponding to the modality with a higher c^m. This is based

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Algorithm 3 KRC-Based Conflict Filtering

Input: Unimodal networks outputs $\mathbf{p}_i^{(m1)}, \mathbf{p}_i^{(m2)}$; threshold ω; modality dependencies c^{m1}, c^{m2}
Output: Conflict flags β^{m1}, β^{m2}
Compute KRC: corr $\leftarrow$ KRC($\mathbf{p}_i^{(m1)}, \mathbf{p}_i^{(m2)}$) (14)
Initialize flags: $\beta^{m1} \leftarrow 1$, $\beta^{m2} \leftarrow 1$
if corr $\leq \omega$ **then**
  if $c^{(m1)} > c^{(m2)}$ **then**
    $\beta^{(m1)} \leftarrow 0$
  else
    $\beta^{(m2)} \leftarrow 0$
  end if
end if
Return: $\beta^{(m1)}, \beta^{(m2)}$

on the intuition that the model should continuously focus on the weaker modality in order to mitigate the imbalance problem in multimodal learning. The filtering flag $\beta_i^{(m1)}$ is defined as follows:

$$\beta_i^{(m1)} = \begin{cases} 1, & \text{if } KRC\left(\mathbf{p}_i^{(m1)}, \mathbf{p}_i^{(m2)}\right) > \omega \ \cup \ c^{(m1)} < c^{(m2)} \\ 0, & \text{otherwise} \end{cases} \tag{15}$$

where $\cup$ denotes the union of event sets and the threshold ω is set to 0 in our experiments. The detailed computational steps are provided in Algorithm 3.

Finally, the overall objective function for the multimodal model is expressed as follows:

$$\mathcal{L} = \mathcal{L}^{(f)} + \lambda \left(\beta^{(m1)} \mathcal{L}_{\text{kd}}^{(m1)} + \beta^{(m2)} \mathcal{L}_{\text{kd}}^{(m2)}\right) \tag{16}$$

where λ is a hyperparameter. In practice, considering the instability of early stage model outputs, λ is initially set to 0 during a warm-up phase and then adjusted to its preset value to ensure stable training.

IV. Experiments and Analyses

A. Experimental Settings

1) Datasets: We conduct experiments on four benchmark dataset settings, based on three publicly available datasets: Houston2013 (HS/DSM) [22], Berlin (HS/SAR) [46], and Augsburg [47]. The Augsburg dataset is split into two subsets according to different modality combinations: Augsburg (HS/SAR) and Augsburg (HS/DSM), where (HS/DSM) and (HS/SAR) denote the combinations of HS and DSM modalities, and HS and SAR modalities, respectively. A detailed description of these datasets can be found in [24].

2) Setting and Metrics: For each dataset, we split the original training data by reserving 10% as a validation set. We adopt three widely used evaluation metrics: overall accuracy (OA), average accuracy (AA), and kappa coefficient (kappa). Higher values indicate better model performance. Detailed calculation methods are provided in [24]. To ensure reproducibility, all sources of randomness (including data splitting, weight initialization, and patch extraction) are controlled by setting a fixed random seed of 5. All experiments are trained for 200 epochs, and the batch size was set to 64. The initial learning rate is set to $5e^{-4}$ and follows a cosine annealing schedule with a warm-up period of 10 epochs. The optimizer used is Adam with a weight decay of $5e^{-4}$. During training, validation is performed every 5 epochs. Accordingly, the modality state weights $\alpha^{(m)}$ defined in (9) are updated at each validation point (every five epochs), consistent with the validation cycle. This keeps the guidance responsive to the model state while avoiding the overhead of per-step purity estimation.

B. Comparison With Other Balanced Learning Methods

1) Quantitative Results: To evaluate the effectiveness of our method, we compare it with several recent state-of-the-art balanced methods, including G-Blending [16], Greedy [20], OGM [17], OGM-GE [18], AGM [21], MMCosine [48], and DRBML [11]. For a fair comparison, all baseline models use the two-branch ResNet18 architecture commonly used in balanced learning work. Detailed network construction is described in Part III of supplementary material. Table I presents the experimental comparisons across the four datasets. It also reports unimodal results to evaluate the baseline performance and the gains achieved through the multimodal fusion. The best overall performance for each dataset is highlighted in bold. The results indicate that our proposed BLMRS method consistently achieves the best performance across almost all metrics and datasets. Out of the 12 total evaluation metrics (4 datasets $\times$ 3 metrics), BLMRS obtains the top score in 11. This consistent improvement over a baseline (joint training) and other specialized balancing methods validates the efficacy of our approach.

2) Qualitative Results: We provide visual results for BLMRS, baseline, and other balanced learning methods for qualitative comparison. In the Houston (HS/DSM) dataset visualization, as shown in Fig. 6, the baseline method misclassifies the parking lots[1] around the stadium and near the highway as residential areas. This misclassification may be due to the spectral similarity of the materials in these areas, specifically the asphalt of the parking lots and the roofs of the residential areas, which exhibit similar reflectance properties in the spectrum. It is worth noting that all balanced learning methods alleviated this problem to some extent, although the OA of some methods (such as MMCosine) was lower than the baseline. Meanwhile, because the lower part of the original aerial image is obscured by shadows, most methods struggle in this region and tend to misclassify the shaded pixels as water or vegetation. The GT reveals a railway running diagonally from the upper left to the lower right of the area, only BLMRS correctly preserves it as a continuous Railway within the shadow zone, whereas the OGM family and AGM introduce gaps, and MMCosine and DRBML omit the segment entirely. For the Augsburg (HS/SAR) dataset, BLMRS shows better water body classification in the lower right region and also improves the classification of industrial areas. For the Augsburg (HS/DSM) dataset, BLMRS not only

[1]We used Google Earth imagery from 2013 to confirm the land was a parking lot, and included the image in Part IV of the supplementary material.

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TABLE I
COMPARISONS OF DIFFERENT BALANCED LEARNING METHODS

Method	Houston(HS/DSM)			Augsburg(HS/DSM)			Berlin(HS/SAR)			Augsburg(HS/SAR)		
	OA(%)	AA(%)	kappa	OA(%)	AA(%)	kappa	OA(%)	AA(%)	kappa	OA(%)	AA(%)	kappa
Only HS	72.09	73.31	0.6982	77.87	50.68	0.6825	70.23	61.55	0.5569	77.87	50.68	0.6825
Only SAR	-	-	-	-	-	-	58.12	31.22	0.3582	82.94	41.06	0.7456
Only DSM	42.33	41.24	0.3774	70.04	35.56	0.5558	-	-	-	-	-	-
Baseline	82.36	82.13	0.8089	82.91	52.92	0.7522	74.90	61.11	0.6092	84.93	50.02	0.7785
Greedy [20]	82.10	83.38	0.8150	83.26	55.69	0.7588	72.99	58.84	0.5859	85.02	50.81	0.7795
G-Blending [16]	81.71	80.70	0.8017	84.45	54.37	0.7730	73.73	61.47	0.6024	84.65	51.07	0.7747
OGM [17]	83.32	83.94	0.8193	82.99	50.96	0.7518	75.09	58.73	0.6037	85.27	52.04	0.7849
OGM-GE [18]	83.41	84.14	0.8201	81.92	51.18	0.7366	74.66	**63.02**	0.6106	83.82	49.47	0.7751
MMCosine [48]	82.23	82.02	0.8078	82.73	55.18	0.7510	73.26	60.70	0.5915	81.57	47.96	0.7301
AGM [21]	83.32	83.66	0.8189	81.94	52.49	0.7380	74.08	60.50	0.6021	84.74	51.79	0.7767
DRBML [11]	83.15	82.99	0.8175	82.84	52.90	0.7516	74.01	62.42	0.6031	84.89	49.91	0.7784
Ours (BLMRS)	**84.97**	**85.52**	**0.8370**	**86.36**	**55.95**	**0.8006**	**76.87**	62.12	**0.6383**	**88.03**	**55.81**	**0.8247**

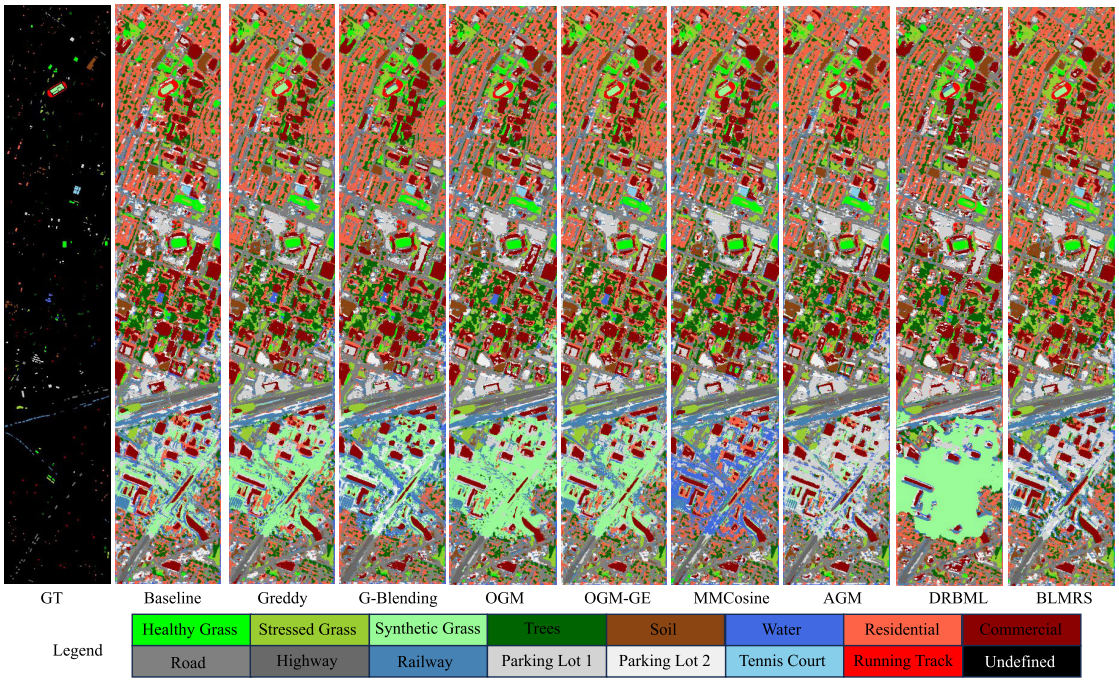


Fig. 6. Visual comparison of BLMRS with other balanced learning methods on the Houston (HS/DSM) dataset.

achieves more accurate water body identification but also reduces false positives in the agricultural zone on the left side, as shown in Fig. 7. In the central region of the Berlin (HS/SAR) dataset, our method improves the baseline performance for large residential areas and reduces the classification noise in the center of the image, as shown in Fig. 8. These qualitative results confirm the effectiveness of the BLMRS method.

C. Adaptability to Different Backbone Architectures

To validate this cross-architecture adaptability, comparative experiments were performed on six representative baseline models from three main categories: 1) CNN-based architectures, including ResNet18 and CO-CNN [3]; 2) Transformer-based architectures, including DSymFuser [23], MFT [5], and ExViT [24]; and 3) Mamba-based architectures, represented by S2Mamba [32]. Results for ResNet18, previously reported in Table I, are omitted here for brevity. As shown in Table II, these architecture-agnostic and dataset-independent improvements confirm the broad applicability of our approach to multimodal remote sensing image classification. Furthermore, the experimental results reveal an interesting phenomenon: our method achieves more pronounced improvements for modal combinations with a greater disparity in information content, such as the fusion of HS and DSM. This indirectly suggests that a larger information gap between modalities is more likely to cause or exacerbate the modal imbalance problem.

D. Adaptability to Fusion Stages and Fusion Methods

A robust balancing learning method should be compatible with various fusion paradigms at different fusion stages of the network. To validate this adaptability for BLMRS, we integrated it with representative methods for data-level, feature-level, and decision-level fusion. Specifically, we tested data-level fusion (denoted by "Data"), feature-level fusion using Summation and the GATE mechanism [49] (replaced concatenation in Table I), and decision-level fusion with the FiLM method [50]. The results are reported in Table III, regardless of the fusion method, applying BLMRS consistently

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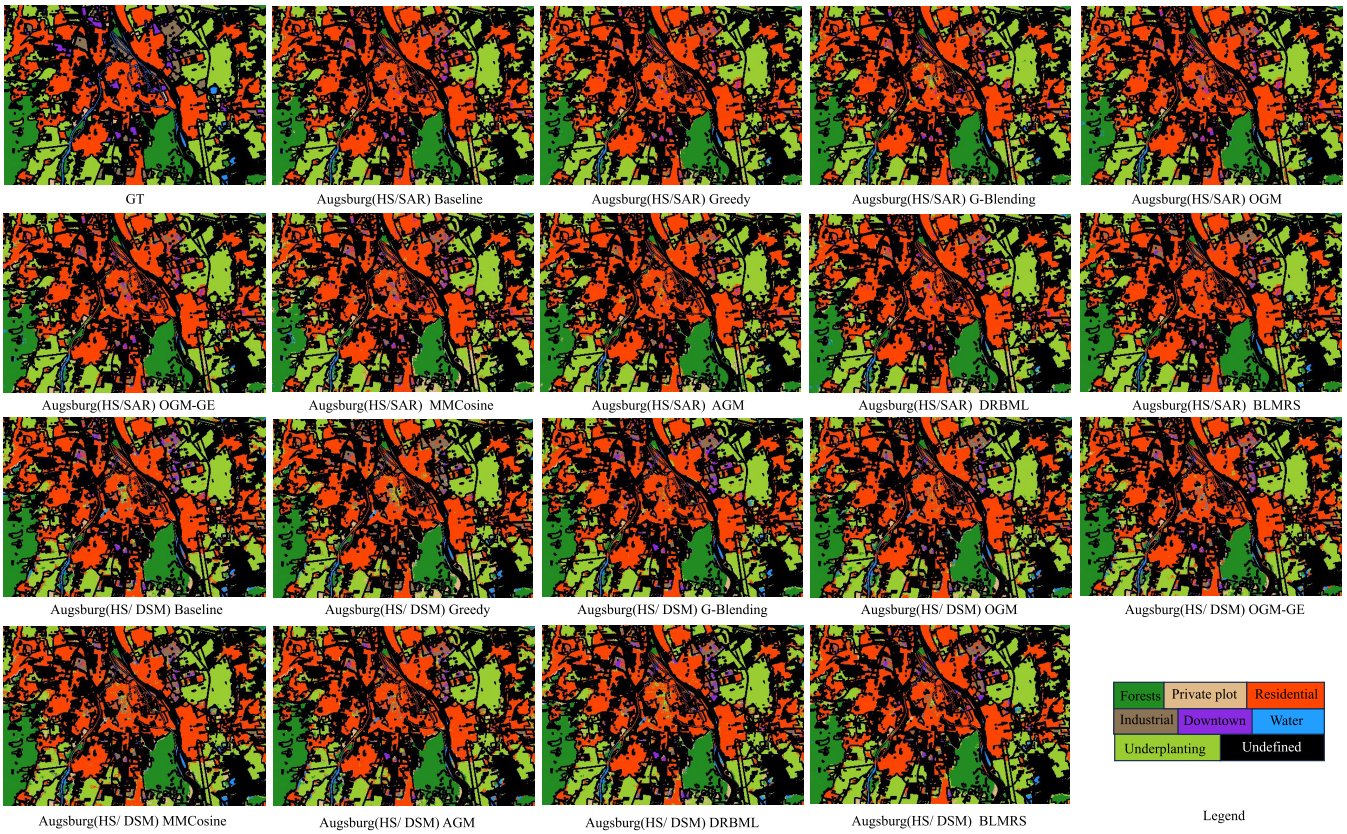


Fig. 7. Visual comparison of BLMRS with other balanced learning methods on the Augsburg (HS/DSM) and Augsburg (HS/SAR) dataset.

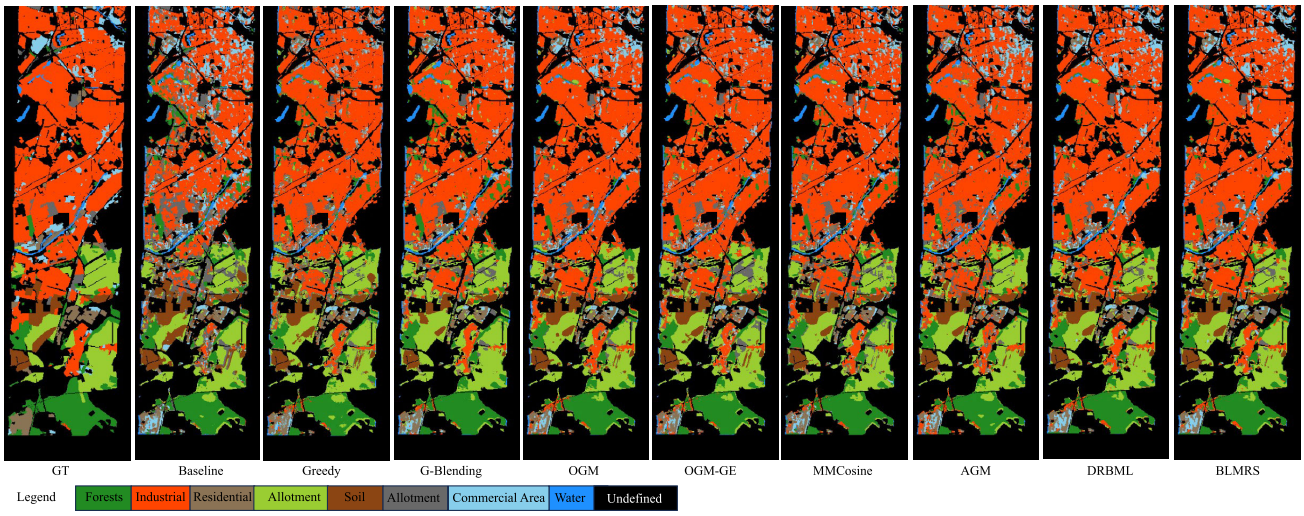


Fig. 8. Visual comparison of BLMRS with other balanced learning methods on the Berlin(HS/SAR) dataset.

led to improved performance. This confirms the effectiveness of BLMRS across various fusion stages and methods. It is worth noting that, compared to feature-level and decision-level fusion, our method brings limited improvements in data-level fusion. This is because the modal imbalance problem becomes more pronounced as unimodal encoding persists [51]. Moreover, the improvement observed with the GATE method is smaller than that achieved with simple concatenation or summation, suggesting that carefully designed fusion mechanisms may help to mitigate modal imbalance to some extent.

E. Ablation Analyses

1) Component Analysis: To dissect the individual contributions of each component within BLMRS, we conducted a comprehensive ablation study. This involves three configurations: 1) removing only the MSA component ("w/o MSA"); 2) removing only the KRC component ("w/o KRC"); and 3) removing both MSA and KRC ("w/o MSA and KRC"). In addition, we replace the KL divergence guidance function with a simpler mean squared error (mse) objective ("repl. mse loss") to assess sensitivity to the guidance function. The quantitative

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TABLE II
ADAPTABILITY ON DIFFERENT NETWORK ARCHITECTURES

Network	Houston(HS/DSM)			Augsburg(HS/DSM)			Berlin(HS/SAR)			Augsburg(HS/SAR)		
	OA(%)	AA(%)	kappa	OA(%)	AA(%)	kappa	OA(%)	AA(%)	kappa	OA(%)	AA(%)	kappa
CO-CNN [3]	87.52	89.61	0.8651	90.33	62.86	0.8619	78.53	56.32	0.6484	91.18	61.77	0.8706
+BLMRS	**89.07**	**90.58**	**0.8816**	**90.79**	**63.34**	**0.8658**	**79.16**	**61.37**	**0.6684**	**91.42**	**63.94**	**0.8759**
MFT [5]	85.50	87.21	0.8427	86.60	63.79	0.8100	73.71	69.46	0.6177	85.91	60.67	0.8004
+BLMRS	**88.88**	**89.89**	**0.8784**	**87.67**	**64.17**	**0.8245**	**74.19**	**69.66**	**0.6258**	**86.36**	**62.45**	**0.8067**
ExViT [24]	89.60	90.89	0.8873	82.73	55.18	0.7510	79.34	54.66	0.6601	87.94	60.34	0.8269
+BLMRS	**90.72**	**91.52**	**0.8993**	**87.80**	**63.55**	**0.8266**	**80.81**	**54.87**	**0.6764**	**88.40**	**61.00**	**0.8337**
DSymFuser [23]	89.10	90.66	0.8820	90.21	67.76	0.8594	74.03	61.11	0.5795	**91.28**	58.88	0.8730
+BLMRS	**90.28**	**91.31**	**0.8946**	**91.05**	**69.11**	**0.8630**	**77.85**	**63.93**	**0.5979**	91.23	**67.92**	**0.8744**
S2Mamba [32]	82.93	85.40	0.8154	90.42	63.70	0.8624	68.35	62.74	0.5512	91.76	67.23	0.8822
+BLMRS	**88.60**	**90.32**	**0.8764**	**91.64**	**65.14**	**0.8799**	**69.12**	**63.39**	**0.5673**	**92.00**	**67.36**	**0.8856**

TABLE III
COMPARISONS OF DIFFERENT FUSION

Fusion Level	Houston(HS/DSM)			Augsburg(HS/SAR)		
	OA(%)	AA(%)	kappa	OA(%)	AA(%)	kappa
Data	82.62	83.16	0.8176	84.29	53.63	0.7720
+BLMRS	**83.35**	**83.39**	**0.8194**	**85.96**	**57.77**	**0.7981**
SUM	83.02	83.24	0.8158	83.82	51.69	0.7645
+BLMRS	**83.79**	**83.47**	**0.8243**	**88.18**	**59.49**	**0.8291**
GATE	83.77	84.90	0.8238	86.39	52.92	0.8021
+BLMRS	**84.91**	**86.13**	**0.8365**	**88.02**	**55.17**	**0.8252**
FiLM	84.68	85.91	0.8338	82.40	51.28	0.7454
+BLMRS	**85.24**	**86.24**	**0.8401**	**86.18**	**55.99**	**0.7994**

TABLE IV
ABLATION STUDY RESULTS

Model	Houston(HS/DSM)			Augsburg(HS/SAR)		
	OA(%)	AA(%)	kappa	OA(%)	AA(%)	kappa
Resnet18+BLMRS	**84.97**	**85.52**	**0.8370**	**88.03**	**55.81**	**0.8247**
w/o.MSA	83.45	83.26	0.8206	87.57	53.73	0.8182
w/o.KRC	82.90	83.17	0.8144	86.71	53.02	0.8048
w/o.MSA&KRC	82.60	82.61	0.8123	85.82	55.66	0.7947
repl. MSE Loss	83.17	83.62	0.8176	85.57	49.75	0.7870
CO-CNN+BLMRS	**89.07**	**90.58**	**0.8816**	**91.42**	**63.94**	**0.8759**
w/o. MSA	88.59	90.20	0.8765	90.67	60.64	0.8633
w/o. KRC	88.19	89.82	0.8722	90.63	59.16	0.8621
w/o.MSA&KRC	87.71	89.38	0.8671	91.07	63.11	0.8698
repl. MSE Loss	87.92	89.69	0.8653	90.77	60.77	0.8647
ExViT+BLMRS	90.72	**91.52**	0.8993	**88.40**	**61.00**	**0.8337**
w/o. MSA	90.13	91.36	0.8931	88.38	60.77	0.8337
w/o. KRC	**90.92**	91.41	**0.9014**	87.33	59.48	0.8185
w/o.MSA&KRC	89.95	91.56	0.8910	88.22	58.62	0.8262
repl. MSE Loss	90.13	91.36	0.8931	87.76	61.50	0.8247
S2Mamba+BLMRS	88.60	90.32	0.8764	**92.00**	67.36	**0.8856**
w/o. MSA	87.20	89.29	0.8614	90.67	65.78	0.8664
w/o. KRC	87.82	89.76	0.8682	88.04	57.83	0.8272
w/o.MSA&KRC	85.28	87.61	0.8406	90.52	62.89	0.8592
repl. MSE Loss	**89.46**	**90.81**	**0.8857**	90.62	**68.05**	0.8663

TABLE V
COMPARISONS OF DIFFERENT HYPERPARAMETER

Network	Houston(HS/DSM)			Augsburg(HS/SAR)		
	OA(%)	AA(%)	kappa	OA(%)	AA(%)	kappa
0	82.36	82.13	0.8089	84.93	50.02	0.7785
0.01	82.56	83.40	0.8112	84.86	51.17	0.7775
0.1	84.64	84.88	0.8333	85.32	49.22	0.7831
1	82.50	83.06	0.8103	85.47	50.33	0.7863
10	83.87	83.86	0.8250	86.04	52.02	0.7950
20	**84.97**	**85.42**	**0.8370**	86.32	52.75	0.7998
30	83.14	83.42	0.8169	**88.03**	**55.81**	**0.8247**
50	82.53	83.08	0.8103	87.13	54.19	0.8116

results, as summarized in Table IV, yield several observations. First, even when both MSA and KRC are removed, the remaining unimodal guidance mechanism alone yields a substantial performance gain over the original baseline (from Table I). This confirms that the core mechanism of using unimodal networks to guide the multimodal model is the primary source of improvement in BLMRS. Second, removing the MSA component also results in a performance decline compared to the full model. This demonstrates the value of dynamically assessing modality states to allocate guidance strength. Third, when the KRC component is removed, performance typically drops. Its absence can lead to significant result instability, as evidenced by the S2Mamba model's performance on the Augsburg (HS/SAR) dataset falling below the baseline. This result underscores the necessity of KRC for resolving conflicting guidance signals. While the occasional, slight performance improvement without KRC suggests a potential regularization effect from signal conflicts [52], we prioritize the consistent and stable results guaranteed by its inclusion. Finally, replacing KL with mse still achieves notable improvements in most cases. This suggests that the effectiveness of our methods is not strictly dependent on a specific function choice.

2) Hyperparameter Analysis: We investigate the sensitivity of BLMRS to the hyperparameter λ, as shown in Table V. Our method consistently outperforms the baseline across a relatively wide range of λ values, indicating that the overall performance is not highly sensitive to the choice of this hyperparameter. However, we observe differences in the optimal value ranges across different datasets. This variation may stem from differences in data distribution, modality characteristics, and noise levels, which in turn affect the model's reliance on the hyperparameter λ. Nevertheless, the broad range of effective λ values demonstrates the robustness of our approach and suggests that it can reduce the reliance on the extensive hyperparameter tuning in real-world applications.

F. Modality Balance Analysis

To evaluate whether BLMRS effectively mitigates the imbalanced multimodal problem, we conduct an analysis that contains quantitative metrics during the training process.

1) Metric-Based Evaluation: We used two metrics to measure the degree of the modal imbalance problem.

a) Modality dependency discrepancy: We modify the modality dependency to $\hat{c}^m = \tau^{f(mm)} - \tau^{f(-m)}$. Unlike (7), this definition becomes strongly negative when ablating modality m increases performance, making the imbalance more salient during evaluation. Based on this modified dependency metric, we define the modality dependency discrepancy to quantify

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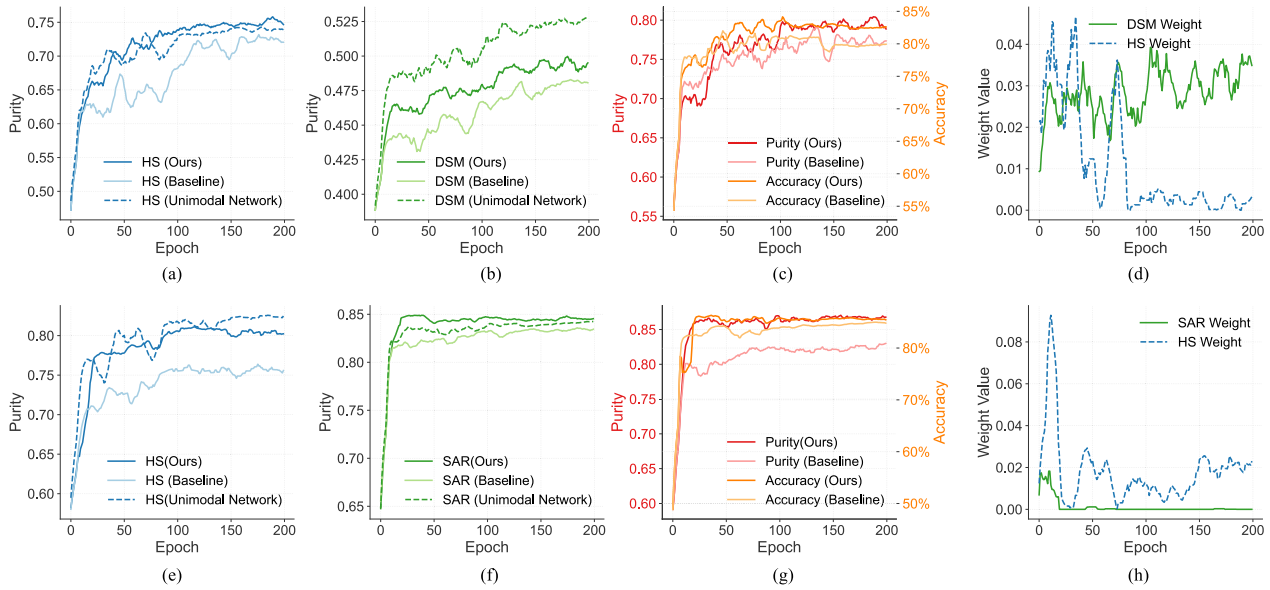


Fig. 9. Purity, OA, and learned MSA weights over training epochs, evaluated on the Houston (HS/DSM) and Augsburg (HS/SAR) test sets. (a) HS purity, (b) DSM purity, (c) multimodal feature purity (left y-axis) and OA (right y-axis), and (d) learned MSA weights. (e)–(h) Corresponding results for Augsburg—(e) HS purity, (f) SAR purity, (g) multimodal feature purity and accuracy, and (h) learned MSA weights.

TABLE VI

COMPARISON OF THE BALANCE DEGREE ACHIEVED BY DIFFERENT BALANCED LEARNING METHODS, WITH THE TOP TWO RESULTS HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	Houston(HS/DSM)		Augsburg(HS/SAR)	
	c_{im}	sh_{im}	c_{im}	sh_{im}
Baseline	0.573	0.291	1.203	0.087
Greedy [20]	0.755	0.313	1.021	0.092
G-Blending [16]	0.806	0.369	1.645	<u>0.064</u>
OGM [17]	0.464	0.262	<u>0.985</u>	0.065
OGM-GE [18]	0.471	0.221	1.045	0.088
MMCosine [48]	0.598	0.275	2.566	**0.026**
AGM [21]	**0.317**	**0.178**	1.238	0.082
DRBML [11]	0.427	0.248	1.793	0.097
Ours (BLMRS)	<u>0.333</u>	<u>0.198</u>	**0.785**	0.077

the degree of modal balance

$$c_{im} = \frac{\left|\hat{c}^{m1} - \hat{c}^{m2}\right|}{max\left(\epsilon, \hat{c}^{m1} + \hat{c}^{m2}\right)} \tag{17}$$

where the absolute value in the numerator ensures that the metric is always nonnegative. A smaller c_{im} indicates that the model's dependence on different modalities is more balanced.

b) Shapley value discrepancy: Building on the cooperative game theory [21], [53], this metric evaluates the contribution of each modality using the Shapley value. The discrepancy is computed as folllows:

$$sh_{im} = \frac{\left|sh^{m1} - sh^{m2}\right|}{max\left(\epsilon, sh^{m1} + sh^{m2}\right)} \tag{18}$$

where, for any modality m, $sh^m = (\tau^{f(mm)} - \tau^{f(-m)} + \tau^{f(m)})/2$, a lower sh_{im} more balanced learning across modalities.

Table VI reports results across datasets. On Houston (HS/DSM), most methods achieve reductions in both c_{im} and sh_{im}, indicating improved modal imbalance problem. But on Augsburg (HS/SAR), several methods yield $c_{im} > 1$, suggesting scenarios where multimodal models underperform relative to unimodal baselines. BLMRS improved the modal imbalance problem across modality combinations, demonstrating the effectiveness of our method.

2) Training Process Analysis: By analyzing the training process, we further verify that BLMRS effectively alleviates the modal imbalance problem. As shown in Fig. 9(a)–(c) and (e)–(g), the evaluation results indicate that DSM is the weaker modality in the Houston (HS/DSM), while HS is the weaker modality in the Augsburg (HS/SAR). In both cases, baseline methods exhibit degraded performance across modalities, with the weaker modality experiencing more severe degradation. In contrast, BLMRS significantly improves the learning quality of the weaker modality, while also maintaining the performance of the dominant modality. This leads to enhanced the feature separability and improved classification accuracy in the multimodal fusion.

In addition, we visualized the evolution of the modal guidance weights learned by the MSA strategy to further verify the reason of BLMRS works. Since the MSA weights are jointly determined by modality dependency and remaining optimization potential, their trends naturally vary with the modality relationship in each dataset. As shown in Fig. 9(d), in the Houston (HS/DSM), DSM serves as the weaker modality and maintains a consistently high weight throughout training, indicating that BLMRS emphasizes that the model should learn from this modality. In contrast, HS, being the dominant modality, receives a relatively low weight. This is consistent with the large information gap between HS and DSM, where DSM provides more limited discriminative information and may suffer from the issue that different land-cover types share identical elevation values. In the Augsburg dataset, as shown in Fig. 9(h), the information gap between the two modalities is smaller, and the difference between the two weights remains at a low level. This indicates that the guidance weights become more balanced when the two modalities show a smaller difference in their relative learning states. Moreover, when the

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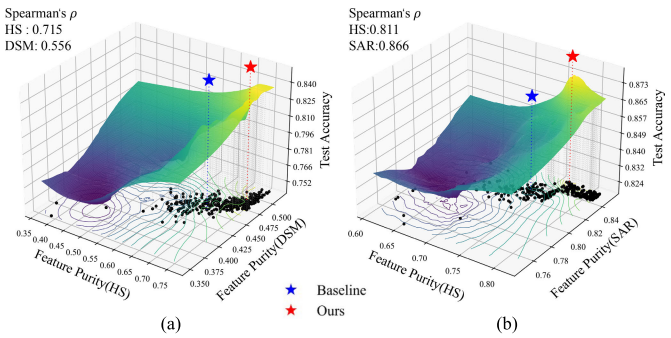


Fig. 10. Visualization of the relationship between unimodal feature purity and classification accuracy. (a) Houston (HS/DSM). (b) Augsburg(HS/SAR).

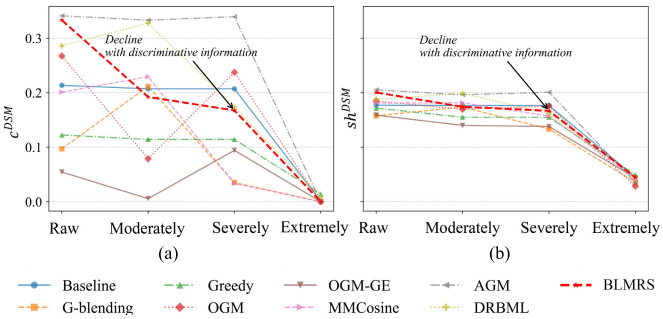


Fig. 11. Evaluation results of different methods at four different DSM elevation resolutions. (a) Evaluation method based model's dependency. (b) Evaluation method based Shapley value.

learning state of a modality in the multimodal model is already close to its unimodal reference, its remaining optimization potential decreases, and the corresponding guidance weight naturally becomes lower. These results show that BLMRS does not adopt a fixed modality preference across datasets, but dynamically adjusts weights based on each modality's relative learning state, thereby mitigating the modal imbalance and improving the overall multimodal performance.

G. Relationship Between Unimodal Feature Separability and Test Accuracy

In this section, we aim to investigate the relationship between unimodal feature separability and multimodal classification accuracy. To this end, we visualize the variations of unimodal feature purity and test accuracy during training on two datasets, as shown in Fig. 10. To enrich the sample distribution, we combine the purity–accuracy data points obtained from both the baseline and our method on the same dataset, providing a more comprehensive illustration of this trend. The figure presents both qualitative and quantitative results, where the quantitative analysis employed Spearman's rank correlation coefficient, a method suitable for assessing monotonic relationships within training data and robust to potential autocorrelation across consecutive training rounds [54]. We additionally report Pearson's linear correlation coefficient as supplementary validation. Complete statistical analysis results are presented in Part VI of the supplementary material. Differences in modality combinations, feature-purity ranges, and class distributions across datasets lead to different surface patterns in the relationship between unimodal feature purity and multimodal classification accuracy. In Houston (HS/DSM), the Spearman correlation between DSM feature purity and multimodal accuracy is lower than that of HS. This is consistent with the limited discriminative information provided by DSM. In Augsburg (HS/SAR), the information gap between the two modalities is smaller, and the feature purity of both modalities shows a stronger and more stable association with the multimodal accuracy. Nevertheless, the overall trend is consistent across datasets: higher unimodal feature separability generally corresponds to better multimodal classification accuracy. This underscores the importance of high-quality unimodal representations for multimodal classification.

H. Experiments on Low-Discriminative Information Modality

To evaluate the effectiveness of BLMRS in fusing low-discriminative information, we conducted experiments on the Houston (HS/DSM) dataset, where the DSM modality is inherently less informative and sometimes assigns identical elevation values to different classes.[2] Leveraging this property, we further degraded discriminative information of DSM and compared BLMRS with other balanced learning methods. Specifically, we define three levels of degradation. Moderately, where the precision of elevation values is reduced to diminish differentiation. Severely, where the precision is further reduced, resulting in greater overlap of values; and extremely, where the elevation values are virtually indistinguishable, and the same elevation value will correspond to all classes. A detailed visualization of the dataset's conflicts under varying degrees of elevation accuracy is provided in Part V of the supplementary material. As shown in Table VII, the experimental results indicate that existing balanced modal learning methods generally perform worse than the baseline as the discriminative information is reduced. In contrast, our proposed method consistently achieves superior performance. To gain deeper insight into this phenomenon, we further examined the modality dependency and contribution of the DSM modality under each method, as modality discriminative information diminished. As illustrated in Fig. 11, the red dashed line corresponds to our method. Notably, ours is the only method in which the model's reliance on the DSM modality consistently decreases with the reduction in modality-discriminative information, as reflected across both base modality dependency[3] and base Shapley value.

I. Comparison With Other Multimodal Classification Algorithms

In this section, we further compare BLMRS with two representative multimodal classification algorithms, quality-aware multimodal fusion (QMF) [55] and predictive dynamic fusion (PDF) [56]. Specifically, QMF dynamically adjusts fusion weights according to the estimated modality quality. The PDF evaluates the contribution of each modality by considering

[2]https://github.com/danfenghong/IEEE_TGRS_ExViT

[3]At this stage, the modality dependency is computed as defined in (7), because in this section we only consider changes in dependencies rather than comparisons with other modal dependencies.

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TABLE VII
EXPERIMENTAL RESULTS WHEN MODALITY INFORMATION IS LIMITED

Method	Moderately			Severely			Extremely		
	OA(%)	AA(%)	kappa	OA(%)	AA(%)	kappa	OA(%)	AA(%)	kappa
Baseline	83.41	83.22	0.8203	79.77	79.89	0.7808	75.62	75.36	0.7361
Greedy [20]	82.29	82.04	0.8080	76.26	75.57	0.7430	75.16	73.88	0.7310
G-Blending [16]	80.55	80.72	0.7894	78.65	78.04	0.7690	74.76	73.58	0.7269
OGM [17]	82.79	82.79	0.8136	79.18	79.39	0.7748	74.79	74.59	0.7272
OGM-GE [18]	82.13	82.17	0.8063	79.27	79.53	0.7756	74.90	73.84	0.7282
MMCosine [48]	81.18	81.85	0.7960	77.67	77.74	0.7582	73.57	73.42	0.7138
AGM [21]	82.85	83.69	0.8142	79.47	79.10	0.7778	74.12	73.82	0.7197
DRBML [11]	82.87	82.92	0.8145	79.84	79.56	0.7818	75.11	74.11	0.7304
Ours (BLMRS)	**84.49**	**84.45**	**0.8318**	**81.18**	**81.51**	**0.7962**	**75.81**	**75.86**	**0.7383**

TABLE VIII
PERFORMANCE COMPARISON WITH MULTIMODAL CLASSIFICATION ALGORITHMS

Dataset	Method	OA(%)	AA(%)	Kappa
Houston (HS/DSM)	Baseline	83.51	83.66	0.8214
	QMF	81.09	82.31	0.7955
	PDF	83.83	83.90	0.8248
	BLMRS	**84.64**	**84.77**	**0.8335**
Augsburg (HS/DSM)	Baseline	81.47	50.11	0.7294
	QMF	84.03	54.39	0.7670
	PDF	83.91	53.21	0.7646
	BLMRS	**85.36**	**57.66**	**0.7884**
Berlin (HS/SAR)	Baseline	73.90	62.75	0.6020
	QMF	73.98	58.71	0.5931
	PDF	74.77	63.08	0.6168
	BLMRS	**75.23**	**64.95**	**0.6230**
Augsburg (HS/SAR)	Baseline	84.57	52.05	0.7749
	QMF	83.72	**57.23**	0.7661
	PDF	85.35	51.96	0.7840
	BLMRS	**86.77**	56.45	**0.8076**

the reliability of unimodal predictions and the consistency of multimodal joint predictions and further calibrates the fusion process through a relative calibration mechanism. Considering that QMF and PDF can both be implemented under a decision-level fusion algorithm, we conduct this experiment under a unified decision-level late fusion ResNet18 setting to ensure fairness. In this setting, each modality is processed by an independent ResNet18 branch and generates the corresponding modality-specific logits output. The fusion operation is performed on the logits outputs of different modalities. The Baseline adopts simple logits addition fusion, BLMRS introduces the proposed balanced learning mechanism on the same basic fusion structure, while QMF and PDF fuse these logits according to their respective dynamic fusion strategies. The results shown in Table VIII demonstrate that BLMRS can still maintain competitive performance, even under late fusion. In addition, we observe that QMF performs poorly on the Houston (HS/DSM) dataset, and the performance improvement of PDF is also relatively limited. We attribute this mainly to the severe low-discriminative-information problem of the DSM modality in this dataset, as mentioned above. Specifically, QMF dynamically assigns weights based on the modality quality, and the ambiguous DSM signals may cause the model to reduce its weight and over-rely on the HS modality, thereby limiting the fusion gain. Although PDF is relatively robust due to its confidence-aware mechanism, when the discriminative cues of DSM are insufficient, the contribution of logits from DSM is still limited. In contrast, BLMRS can dynamically estimate the modality learning state and adjust the learning preference accordingly. Therefore, it is less affected by the above bottleneck and achieves a more robust performance improvement.

TABLE IX
COMPARISON ON THE MULTIMODAL SCENE CLASSIFICATION TASK WITH AND WITHOUT BLMRS

Dataset	Methods	F1(%)	OA(%)	AA(%)	kappa
ADVANCE (Audio/RGB)	Baseline	95.24	95.37	94.60	0.9477
	With BLMRS	**95.97**	**96.65**	**95.46**	**0.9622**
EuroSAT (RGB/SAR)	Baseline	95.91	96.04	95.91	0.9559
	With BLMRS	**96.59**	**96.69**	**96.55**	**0.9631**

V. DISCUSSION

A. Application of BLMRS to Other Classification Tasks

To further assess the adaptability of the proposed BLMRS method beyond pixel-level remote sensing analysis, we evaluate its performance on two tasks that differ in both data modality and task granularity: image-level multimodal scene classification and sequence-level driver state recognition. These tasks introduce modalities not previously covered, including RGB, audio, infrared (IR), and depth data, while also involving 3-D feature extraction models for spatiotemporal modeling, unlike the 2-D architectures used previously. The training strategy and other hyperparameters in this section remained consistent with previous experiments. We introduced the F1-score to provide a more objective assessment of the model's ability to recognize minority classes.

For the multimodal remote sensing scene classification task, experiments on the ADVANCE (Audio/RGB) dataset [57] followed the protocol proposed in [58]. This dataset provides paired RGB image frames and corresponding audio tracks. For the EuroSAT (RGB/SAR) dataset, we constructed multimodal paired data by combining EuroSAT-RGB [59] and EuroSAT-SAR [60]. The baseline model used in these experiments was consistent with Table I. As shown in Table IX, the inclusion of the BLMRS module led to performance improvements across all four evaluation metrics.

In the driver state classification task, unlike prior work [61] which only considered a single abnormal state category, we evaluated the effectiveness of the proposed BLMRS method on a more fine-grained abnormal state classification setting. Given that this task is based on video data from the driver anomaly detection (DAD) dataset [62], the data preprocessing pipeline and baseline model configuration followed the setup

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TABLE X
CLASSIFICATION RESULTS ON THE DAD VIDEO DATASET WITH AND WITHOUT BLMRS

Viewpoint	Methods	F1(%)	OA(%)	AA(%)	kappa
Front View (IR/Depth)	Baseline	35.72	70.13	39.16	0.3482
	With BLMRS	**41.68**	**77.19**	**41.64**	**0.4171**
Top View (IR/Depth)	Baseline	30.75	72.02	35.21	0.3739
	With BLMRS	**38.44**	**79.52**	**37.19**	**0.4441**

TABLE XI
CLASSIFICATION RESULTS ON THE AUGSBURG (HS/SAR/DSM)

Methods	OA(%)	AA(%)	kappa
Baseline	84.79	51.54	0.7776
G-Blend	87.03	48.74	0.8079
AGM*	85.27	50.27	0.7854
DRBML	85.90	50.93	0.7934
Ours(BLMRS)	**88.59**	**56.10**	**0.8325**

described in [61]. Experiments were conducted by fusing IR and depth modalities captured from two different in-cabin viewpoints: the front view and the top view. The results in Table X demonstrate that the proposed method consistently achieves superior performance across different viewpoints, further confirming its robustness and adaptability in complex multimodal scenarios.

B. Application of BLMRS to Three Modalities Fusion Scenario

The trimodal extension of BLMRS largely follows the bimodal design, with only slight modifications where necessary. The following describes how MSA weighting, unimodal guidance, and KRC-based conflict filtering operate under three modalities.

For the MSA strategy, the process for estimating guidance weights is identical to the bimodal approach. We consider three modalities $m \in \{m1, m2, m3\}$, and calculate their respective dependency scores $c^{(m)}$ and optimization potentials $d^{(m)}$. The final modality state weights $\alpha^{(m)}$ are then obtained using (9). Similarly, the unimodal guidance mechanism is extended by instantiating and training three distinct unimodal networks, one corresponding to each modality. The learning of the multimodal network is steered by the same guidance objective function, now aggregating signals from all three guides. For conflict filtering, the KRC-based consistency filtering needs to be adjusted accordingly. Specifically, the two modalities with the lowest modality dependency are first identified, and the one with the higher modality state weight is regarded as the modality that most urgently needs optimization. Subsequently, the KRC values between this modality and the other modalities are computed, and according to a predefined threshold, the guidance signals from other modalities that are inconsistent with it are filtered out. This modification aims, by prioritizing the retention of guidance information from the modality that requires more reinforcement, to mitigate the negative impact caused by modality imbalance to the greatest extent in a trimodal scenario, where two weak modalities may coexist.

To illustrate, suppose that modalities $m1$ and $m3$ exhibit lower dependency, i.e., $c^{(m2)} = \max\{c^{(m1)}, c^{(m2)}, c^{(m3)}\}$, indicating that $m2$ is the dominant modality, while $m1$ and $m3$ are relatively weaker. If $m1$ has the larger modality state weight among the two weaker modalities, i.e., $\alpha^{(m1)} > \alpha^{(m3)}$, it is selected as the modality requiring adjustment. We then compute the pairwise KRC values KRC$(m1, m2)$ and KRC$(m1, m3)$. For any modality $mj \in \{m2, m3\}$, whose KRC score with $m1$ satisfies KRC$(m1, mj) < \omega$, where $\omega = 0$, the corresponding guidance signals are filtered out. This procedure ensures that only consistent and supportive guidance is retained to stabilize the learning of the selected modality $m1$. We report the trimodal experimental results on the Augsburg (HS/SAR/DSM) dataset in Table XI, comparing BLMRS against baseline and other balanced multimodal learning strategies capable of handling three modalities. The results demonstrate the superiority of BLMRS. (The * denotes that the method does not support the trimodal setting; we extend it following the original idea to obtain this result.) To provide comprehensive information, Part VII of the supplementary material presents the feature purity, test accuracy, and the modality weights during training under the trimodal setting of BLMRS.

TABLE XII
COMPARISON OF TRAINING AND INFERENCE COSTS

Methods	Houston(HS/DSM)			Augsburg(HS/SAR)		
	Train Time	GPU Memory	Infer. Time	Train Time	GPU Memory	Infer. Time
Baseline	1.28	1013	2.42	0.60	969	15.54
Greddy	1.85	1013	2.42	0.75	969	15.54
G-Blending	3.02	1107	2.42	1.17	1061	15.54
OGM	1.70	1021	2.42	0.70	973	15.54
OGM-GE	1.74	1021	2.42	0.74	973	15.54
MMCosine	1.31	1013	2.42	0.61	969	15.54
AGM	1.90	1015	2.42	0.76	969	15.54
DRBML	1.75	1023	2.42	0.74	979	15.54
Ours(BLMRS)	3.77	1177	2.42	1.61	1133	15.54

C. Limitation of BLMRS

Although BLMRS achieves superior performance, its training cost is higher than that of other methods. The increased training time mainly arises from the per-sample purity estimation in the MSA module and the classwise pairwise probability computation of KRC, while the additional GPU memory consumption results from the parameters of the extra unimodal networks. It should be noted that balanced multimodal learning is a training-stage strategy, and all related operations are applied exclusively during training. Therefore, none of the methods incur additional inference overhead upon deployment. Table XII reports the training time (in seconds, including validation), GPU memory usage (in megabytes), and inference time (in seconds) of BLMRS, showing that it requires more computational resources than existing methods. Moreover, the computational paradigm of KRC is not well-suited for dense prediction tasks, which limits its scalability. Future work can focus on introducing lightweight internal imbalance adjustment mechanisms to reduce reliance on external reference networks, as well as exploring more efficient KRC approximation methods to enhance the scalability and applicability of the approach.

VI. CONCLUSION

In this study, we pointed out that due to the inherent lack of discriminative information in some remote sensing

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modalities, an incorrect focus on learning such modalities can have a negative impact on multimodal models. In addition, remote sensing scenarios require a more flexible approach to accommodate diverse network architectures. To this end, we propose the BLMRS, which leverages an MSA strategy, an architecture-agnostic unimodal network guide mechanism, and a KRC-based conflict filtering to adjust the multimodal model's learning preferences to different modalities. Extensive experiments across various datasets and network architectures demonstrate that BLMRS consistently enhances classification performance and exhibits adaptability, outperforming several state-of-the-art methods, particularly under scenarios with reduced modality discriminative information. Furthermore, preliminary results on the multimodal scene classification and driver state classification suggest that our approach holds promise for broader applications in other multimodal classification tasks. We hope this work serves as an inspiration for alleviating modal imbalance in remote sensing and fosters further research on complex scenarios.

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